

Strategy Room

An F1 pit-stop strategy simulator — project overview

What is this, in plain words?

Strategy Room is a website where you pick a real Formula 1 race from the past, and then "replay" it with a different pit-stop strategy than what actually happened. For example: what if the driver had pitted for new tires on lap 15 instead of lap 18? The site calculates how that change would likely have affected the race, and shows you whether the driver would have finished faster or slower, and in what position.

Think of it like a "what if" machine for race strategy — the same kind of thinking real F1 race engineers do on the pit wall, except here anyone can try it out for any past race.

Why this is interesting

- It is not just a stats page showing old race results — it actually calculates new outcomes based on your choices.
- It uses real historical race data, so the comparisons are grounded in what actually happened.
- It accounts for randomness — like safety cars — the same way real strategists have to plan around unpredictable events.

How it works, step by step

- 1. Pick a race** — Choose any past Grand Prix from a list — for example, 2024 Monza.
- 2. Pick a driver and change the strategy** — Choose a driver, then drag markers on a simple timeline to decide when they pit for new tires and which tire type they use.
- 3. Run the simulation** — The website calculates, lap by lap, how fast the car would go on the tires you chose, including a time penalty for the pit stop itself.
- 4. See the result** — You get a side-by-side comparison: your imagined strategy vs. what actually happened in the real race, including estimated finishing time and position.

How the calculation is done (the engine)

Tires get slower the longer they are used, and different tire types (soft, medium, hard) wear out at different rates. The system uses this to work out lap times. A pit stop itself costs around 20-25 seconds of lost time. Because real races have random events like safety cars that can help or hurt a strategy, the site does not just run the calculation once — it runs it around 500 times with slightly different random conditions, and shows the range of realistic outcomes rather than a single guess. This is the same approach (called a Monte Carlo simulation) used in fields like finance and weather forecasting to handle uncertainty.

What I'm building it with

Part	What it does	Tools used
Frontend (what you see and click)	The web pages, the pit-stop timeline you drag, the charts comparing strategies, and the animations that make it feel smooth and responsive.	React (page structure) Tailwind CSS (styling) Framer Motion (animations) Recharts (charts)
Backend (the engine behind the scenes)	Fetches real race data, stores it, and runs the tire-wear and pit-stop calculations whenever you hit “run simulation.”	FastAPI (Python web server) PostgreSQL (database)
Data source	Where the real historical race data (lap times, pit stops, results) comes from.	Jolpica API (free, public F1 statistics database)

How it will turn out

The end result is a working website — not just a design or a demo — that anyone can open, pick a real race, redesign the pit strategy, and immediately see how that choice would have played out. It will feel closer to a simple game or interactive tool than a spreadsheet or dashboard: drag a marker, press a button, watch the result animate in.

For anyone I explain this to, the one-line version is: **"It's a website where you can rewrite F1 pit-stop decisions from real races and see if your call would have been faster."**

What it's actually useful for, right now

This is a personal project, not a commercial product, so it's worth being clear about what it is genuinely useful for at this stage:

- A skill demonstration: it shows the ability to build a full working website end to end — database, backend calculations, and an interactive frontend — which is the main point of building it right now.
- A learning and exploration tool for F1 fans: anyone can try out “what if” strategy calls on real past races and get a genuine, data-backed answer instead of just guessing.
- A talking piece for internships and interviews: a live, working product is easier to explain and more convincing than a resume line, especially since it involves real data, a backend, and a database.
- A base to build on later: the same approach (historical data plus a what-if simulation) could extend to other sports or even other decision-based scenarios in the future.

What it is **not**, at least for now: it's not a tool used by any real F1 team, and it doesn't predict future races — it only replays and adjusts strategies for races that already happened.